Privacy – Enhanced Synthetic Data Generation for Autonomous Vehicle Training

R26-CS-010

Individual Thesis

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Individual Component: Utility Evaluation

Group ID: R26-CS-010

BSc (Hons) Degree in Information Technology Specialized in Cyber Security

Department of Computer Systems Engineering

Sri Lanka Institute of Information Technology

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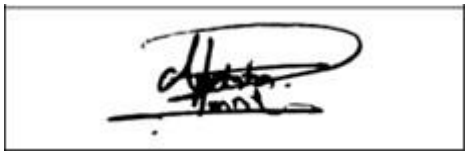
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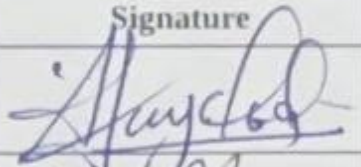
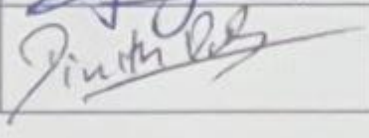
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Supervisor certification. The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

Supervisor certification. The above candidate carried out this research for the bachelor's degree thesis under my supervision.

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Lastly, I would like to thank the open source community consisting of Ultralytics, PyTorch, CARLA and torchmetrics for the development of the tools used in this evaluation process.

Abstract

Autonomous vehicle perception systems need high amounts of labeled driving imagery that has been traditionally collected in real footages with the presence of recognizable entities such as pedestrian faces or car license plates. There is another way to create synthetic data for privacy preserving data training. However, privacy preserving does not suffice to demonstrate the applicability of generated data. In this report, we describe the utility evaluation module for a four staged privacy preserving synthetic data pipeline, in which the differential privacy (Laplace mechanism) calibration stage preps CARLA simulator for generation of synthetic driving footage. The security validated evaluation process includes the following steps: decryption, integrity verification, data poisoning attack, exploratory data analysis and YOLOv8 object detection model training and evaluation of five towns, 10000 frames synthetic dataset. We provide Map scores for overall dataset as well as mAP scores per class for the test split of the dataset (mAP 50 – 95 = 0.2273).Comparison of performance between models trained on synthetic data and real data on the real data (KITTI benchmark) is provided, with sim to real gap being apparent as the former achieves only 10.46% of the performance of the latter (mAP50=0.0889 vs. 0.8500). A Fréchet Inception Distance (FID) score of 115.99 is reported for a quantitative evaluation of the statistical similarity of the generated image distribution with the real one (FID = 115.99). It can be concluded form the results that the privacy preserving synthetic data can be used to train a useful AV perception model, where the difference in performance is measured based on the dataset composition and not the failure of the model.

Keywords: Privacy Preserving Synthetic Data, CARLA Simulator, YOLOv8, Mean Average Precision, Fréchet Inception Distance

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List of Abbreviations

Abbreviation	Full Form
AES	Advanced Encryption Standard
AP	Average Precision
AUC	Area Under the Curve
AV	Autonomous Vehicle
CARLA	Car Learning to Act (open source driving simulator)
DP	Differential Privacy
DP-SGD	Differentially Private Stochastic Gradient Descent
FID	Fréchet Inception Distance
GDPR	General Data Protection Regulation
GPU	Graphics Processing Unit
IoU	Intersection over Union
KITTI	Karlsruhe Institute of Technology and Toyota Technological Institute (dataset)
mAP	mean Average Precision
PDPA	Personal Data Protection Act (Sri Lanka)
SHA-256	Secure Hash Algorithm 256 bit
YOLO	You Only Look Once (object detection architecture)

Table 1: List of Abbreviations Used in this Report

1. Introduction

1.1 Overall Project Context

The Autonomous Vehicle (AV) perception algorithms require vast amounts of manually labeled images that are capable of identifying road users such as pedestrians, cyclists and other road users. Such data sets have been gathered from videos captured in real life conditions; therefore, such images include personal information such as pedestrian faces and license plate numbers of cars. Utilizing such data to build machine learning algorithms brings about legal issues in terms of data privacy laws such as General Data Protection Regulation (**GDPR)** [11] and Personal Data Protection Act (**PDPA)** of Sri Lanka [12].

Simulation of driving data has also been suggested as a potential approach that would substitute data collection by simulating data generation. However, such an approach would not solve the problem of privacy since, in order to configure the simulation environment, data collected in real world driving should be used (such as traffic density and speed). Therefore, the problem of privacy would merely shift to another stage in the process of data processing.

This is tackled through the group study by use of four stages privacy preserving synthetic data pipeline, as known in table below.

Stage	**Owner**	**What it does**
1. Privacy Calibration	IT22309556	Adds mathematical noise (the Laplace mechanism) to real world statistics before they reach the simulator, and applies DP-SGD during downstream model training.
2. Dataset Generation	IT22541284	Configures the CARLA simulator using the privatized statistics and generates synthetic driving footage across five towns.
3. Utility Evaluation	IT22110220 (this report)	Measures whether the resulting synthetic dataset is actually good enough to train a real AV perception model.
4. Risk and Compliance	IT22066916	Attacks the pipeline and maps the implemented controls to relevant GDPR articles.

1.2 Individual Component Context

Thesis work includes the Utility Evaluation Component which is the third step in the process. It takes in the encrypted synthetic data generated by the Dataset Generation component and delivers quantitative utility results (detection performance, image realism, and label integrity). It provides the poisoning filtered labels and the detection results used in the analysis done by the Privacy Integration and Risk Compliance components.

Utility Evaluation component addresses an important question which the other three steps cannot address: Is the dataset useful after applying privacy protection measures at the earlier stages of the process? Without answering this question, the project can provide privacy but not usability.

1.3 Problem Statement

Privacy becomes a simple problem to solve when usefulness is totally ignored, as a dataset that only consists of noise can satisfy very good privacy properties, while at the same time not being useful at all to train any models. The Utility Evaluation module solves one of the key problems of privacy preserving data synthesis studies. Privacy properties alone do not create any practical usefulness. In the absence of utility measurements, the team's overall experiment could boast privacy guarantees, yet would be unable to assert the practical usefulness of the synthetic data produced, ignoring the "utility" part of the privacy utility trade – off.

An additional challenge is that the privacy preserving synthetic dataset used for evaluation purposes was created with a single calibration privacy value (epsilon = 1.8) because the architecture of the pipeline produces the privatized values of configuration parameters before generating scenes and not the privatized dataset directly. In this report, I will discuss how Utility Evaluation can be carried out in spite of this limitation.

1.4 Research Gap

Current research on privacy preserving datasets for AVs can be viewed from two different perspectives: the methods for providing privacy (such as differential privacy in input level before the simulator [1]) and the level of realism in the images (such as FID [5]). However, it is evident from the section 2.5 that both guarantees are not enough to make sure that a dataset will generate a working perception model, as an empty set of random noises fulfills the first requirement but not the second.

As seen from the literature discussed in Chapter 2, there are four remaining gaps for building a pipeline, where a simulator is calibrated by DP statistics:

- **Missing utility evidence.** There is no measurement of how an object detector trained only using DP calibrated CARLA data is effective, especially when it is applied to images from real world imagery.
- **Aggregate metrics hide class level failure.** Overall mAP metric might mask really poor performance of the underrepresented and small object detection classes like pedestrians and cyclist [7], the most critical in terms of safety of all.
- **Unknown relationship between realism and utility.** It is not known whether FID scores predict the effectiveness of detection in the case of this type of dataset.
- **Unaudited external data.** Labeling done through another component in the pipeline is considered valid implicitly yet label level data poisoning is possible without changing an image [9].

This individual metric compensates for the shortcoming in the mAP comparison by carrying out an mAP comparison using four different tests on different domains compared to an independent real world benchmark (KITTI), an FID metric, class, and size report and label integrity test on the detector. All these form the utility side of the privacy utility tradeoff of the process of the group.

1.5 Research Aim

An empirical test of the efficacy of training a fully functional autonomous vehicle object detection system with privacy preserving synthetic driving data that is calibrated using differential privacy parameters is required as compared to training the same system using real world driving data.

1.6 Objectives

1.6.1 Main Objective

The primary goal of this study is to measure, through rigorous experimentation, whether a privacy preserving synthetic dataset generated by the group's CARLA based pipeline is capable of training an object detection model (car, pedestrian and cyclist) to a standard comparable with a model trained on real world data and to quantify the resulting utility cost in defensible, reproducible terms.

1.6.2 Specific Objectives

1. To securely decrypt and verify the integrity of the received synthetic dataset using AES 256 decryption and SHA 256 manifest verification.
2. To statistically explore the dataset's class distribution and object size composition prior to training, in order to anticipate and explain downstream performance patterns.
3. To implement and empirically validate a data poisoning detection mechanism that audits label integrity before training, addressing the security component required of this project.
4. To convert the dataset into YOLO training format and train a YOLOv8n object detection model on the synthetic data, reporting overall and per class mean Average Precision (mAP) on a held out test split.
5. To establish an independent real world baseline (the KITTI benchmark) and conduct a four experiment cross-domain evaluation to quantify the sim to real utility gap.
6. To compute the Fréchet Inception Distance (FID) between the synthetic and real image distributions as a complementary, task-independent measure of dataset realism.

1.7 Research questions or hypotheses

These research questions guide the work:

i. **RQ1.** Can an object detection model (YOLOv8), which is exclusively trained on the privacy preserving synthetic dataset, be able to detect cars, pedestrians and cyclists in the held out synthetic images?

ii. **RQ2.** How much of the detection performance does the model trained on synthetic images preserve as opposed to the model trained on real images in the real world imagery (KITTI)?

iii. **RQ3.** How is the preservation of the utility distributed across different object categories and object sizes?

iv. **RQ4.** Is there any domain gap between CARLA rendering and real world imagery for bidirectional transfer learning?

v. **RQ5.** Does the level of detection utility depend on the image level realism score (FID)?

vi. **RQ6.** Is the dataset generated free of label manipulation?

Three hypotheses of research can be formulated corresponding to these questions. **H1:** there is enough data for training a good detector since the results of classification on a test dataset of artificial images are undoubtedly better than pure chance. **H2:** the performance on real world images for a detector trained on artificial data will be worse than for a detector trained on real world data. **H3:** this performance will not be the same for all classes and will depend on their occurrence and size.

1.8 Scope limitations and assumptions

Scope: - This report concerns itself with stage 3 (Utility Evaluation) of the four stages pipeline followed by the group. This includes the usage of the 10,000 frame synthetic CARLA dataset, which includes five towns and is of dimension 800x600 with RGB frames and YOLO labels, three object categories (cars, pedestrians and cyclists) YOLOv8n detector, KITTI 2D object Detection benchmark as real world baseline, mAP50 and mAP50-95 as detection metrics, FID as the realism metric and label integrity audit. Stage 1 (Privacy Calibration and DP – SGD) and Stage 2 (Dataset Generation) and Stage 4 (Risk & Compliance) have been carried out by other members of the group.

Limitation: - The data set is made only for a single privacy value of epsilon = 1.8, which cannot provide any privacy versus utility graph with varying privacy budgets. CARLA (800x600, 4:3) and KITTI (approximately 1242x375) images have different resolutions and aspect ratios, which leads to some part of the simulation real transfer problem. Experiments were performed on a CPU (Intel Core i5 – 8250U), limiting this experiment to the small version of YOLOv8. There was also a limit on the training budget, which had to be 50 epochs. Each result shown here is from one trained model for each domain with a fixed seed of 42.

Assumption: - (i) KITTI is an adequate proxy for realistic driving images and as such does not suffer from circularity due to the lack of correlation with the statistics provided by Waymo. **(ii)** An unencrypted file with a matching SHA – 256 hash to the manifest file is an exact copy of the one generated during the generation phase. **(iii)** The provided labels are the ground truth unless the results of poisoning audit indicate anything different. **(iv)** Mapping of KITTI categories into three categories of this project is semantically equivalent to the synthetic classification system. **(v)** Differential privacy in stage one is guaranteed.

1.9 Individual contribution

My personal contribution to the group project is the design, development and execution of the entire utility evaluation stage, which includes:

- Decryption module (AES - 256 – CBC) with SHA 256 manifest validation of the downloaded dataset (40,000 images, five towns)
- Data exploration for class balance, empty images and objects sizes and per class downstream performance explanation
- Data poisoning detection algorithm (structural and statistical validation) and proof of concept injection
- Synthesis and conversion scripts for preparing the synthetic dataset and KITTI baseline in the same YOLO format with split in 70/15/15 with the same seed
- Training of two YOLOv8n models (with synthetic and KITTI datasets) using the same hyper parameters, in four different domain evaluation modules
- FID computation module and discussion of mismatching between FID and mAP
- Conclusions that give information on privacy cost at the data level for the privacy utility trade off discussion of the group

The other modules that the rest of the group members have (Privacy Calibration, Dataset Generation, Risk and Compliance) are briefly explained in order to understand the connection between this module and the others.

1.10 Thesis structure

This thesis will be structured in the following manner.

Chapter 2 (Literature Review and Related Works), topics to be explored include risks of privacy in AV training dataset, Differential Privacy and Laplace mechanism, data poisoning, CARLA synthetic data, YOLO and mAP evaluation and FID and ends with gap synthesis.

Chapter 3 (Research Methodology) the method employed, data set used, training carried out, cross domain and FID method, metrics utilized, exploratory analysis, security and ethics issues and reproducibility will be covered.

Chapter 4 (Product and Solution Design) will cover the requirements, role of the component in the entire system and solution design.

Chapter 5 (Implementation, Integration and Testing) will cover the development environment, implementation issues, integration with other components, test results and solution operation.

Chapter 6 (Results and validation) covers results of Detection, FID, Poisoning Audit, Integration & Conclusion that are drawn from them.

Chapter 7 (Discussion) deals with results & interpretation thereof, comparison with literature, research / product contribution, limitations & implications.

Chapter 8 (Conclusion and Future Directions) covers thesis summary, accomplishment of each objective of the work, individual contributions & future works. The report ends with references, glossary & two appendices.

2. Literature Review and Related Work

2.1 Review method and scope

Due to an increase in the use of large annotated datasets in autonomous car research, there is an urgent need to give attention to the privacy concerns that arise in relation to the use of real world driving videos. This chapter discusses the review of the relevant literature on the issue of regulation of driving data privacy, differential privacy and synthetic data generation in self-driving cars, object detection and data poisoning. After that the research gap will be defined.

2.2 Domain and theoretical foundations

2.2.1 Privacy risk in Autonomous Vehicle Training Data

Driving data will always have personally identifiable information such as faces of pedestrians and license plates of cars. Processing of data for the training of AI models is governed by various legislative documents such as the GDPR (General Data Protection Regulation) in Europe [11] and Personal Data Protection Act in Sri Lanka [12].Therefore there is considerable liability risk associated with collection and processing of the driving data; hence, the need for privacy preserving techniques.

2.2.2 Differential Privacy and the Laplace Mechanism

The idea of differential privacy provides a mathematical formulation such that when information regarding the data set is released, inference regarding any particular record within the dataset is only minimally impacted [1]. One of the important techniques used in differential privacy includes the use of Laplace mechanism, where the techniques operates on the Principles of addition of statistical noise prior to the release of the aggregate values. This takes place within the larger context of the group pipeline, where traffic statistics from the real world are first adjusted prior to simulation setup.

2.3 Related research

2.3.1 Data Poisoning in Machine Learning Pipelines

The learning pipelines where the data that is collected from some other third party is used are susceptible to attacks due to poisoning the data by means of poisoned labels in such a way that the learning model becomes corrupt without making any changes to the actual images [9]. The altered bounding boxes will train the model to detect objects at wrong locations and altered labels will train the model to detect objects that don’t even exist. The threat as described above is highly pertinent to the multi component pipeline approach followed in our case study.

2.4 Existing systems products and techniques

2.4.1 Synthetic Data Generation for Autonomous Driving (CARLA)

The CARLA simulator is an open source simulation platform built for self-driving car development, offering the possibility of traffic, weather and sensor configurations [2]. The use of the synthetic data that can be generated via simulators such as CARLA has been proposed as a solution for ensuring privacy preservation when compared to collecting real world data in which no people take part. However, since the real world statistics have to be used for making the simulations more realistic, privacy concerns are only delayed until the calibration phase of the simulator as shown in this report.

2.4.2 Object Detection Evaluation with YOLO and mAP

The real time object detection algorithm that is used most often include You Only Look Once (YOLO) models which are used to solve the problem of object detection as a sinlge regression problem in the whole [3]. The detection accuracy of the algorithm is typically estimated by mean Average Precision (mAP) values, which is a common measure of efficiency for object detection in the benchmarking datasets such as the Microsoft COCO dataset [6]. It is recommended to perform class by class evaluation in order to avoid hiding of low detection accuracy of some less common classes by using mAP metrcis [7].

2.4.3 Evaluating Synthetic Data Realism with FID

The Fréchet Inception Distance (FID) is among the metrics which have become popularly employed in measuring the similarity of statistics between two datasets with respect to feature representations obtained from a pre trained Inception V3 model [5]. The lower the value of FID, the greater the similarity between the two distributions of images. FID has found its application in generative models as a task independent metric to the task specific metrics, and thus can be justified for inclusion alongside mAP.

2.5 Gap synthesis and conceptual framework

Based on the literature reviewed above, it can be concluded that **(i)** differential privacy algorithms may be used upstream from the production of synthetic datasets. However, it means that the risk of privacy has been transferred rather than removed. **(ii)** The performance of object detection should be measured for each class, rather than just overall, because there is an issue of class imbalance; and **(iii)** FID is a meaningful metric to measure realism regardless of the particular task at hand. However, it does not provide evidence regarding whether a specific privacy – preserving synthetic driving dataset will have sufficient utility for training a practical perception model compared to real world data. Neither does it use a combination of mAP and FID metrics with an integrity check of the data.

3. Research Methodology

3.1 Research Design

This chapter provides information on the approach used in designing, development and evaluation of the utility of the privacy preserving synthetic dataset. The approach used in this study focuses on the whole process of evaluating the utility of the data from manipulation of the data to training of models in the respective domains.

3.1.1 Research Approach

An empirical experimental approach was used to perform the research. The assessment was performed through the comparison of detection efficiency in four pairs of train / test data sets constructed in a systematic way, along with the distribution similarity metric (FID) and a security validation technique (data poisoning detection). The steps of the methodology comprised:

- Environment setup and checking
- Securely decrypting and verifying the integrity of the received dataset
- Exploratory data analysis (class distribution, object size distribution)
- Detecting data poisoning, including testing the detector empirically
- Converting the dataset to YOLO training format
- Training and evaluating a YOLOv8n model on the synthetic dataset
- Collecting and converting an independent real world dataset (KITTI)
- Cross domain evaluation across four train/test combinations
- Computation of the Fréchet Inception Distance (FID)
- Result aggregation and analysis

3.2 Data, participants or study materials

No human participants took part in this study. The materials were a synthetic dataset received from Dataset Generation and a public real world benchmark.

3.2.1 Synthetic dataset

Synthetic dataset was created using Dataset Generation module and CARLA simulator with calibrated privacy parameter values (total **ε = 1.8** on seven statistical features based on Waymo Open Dataset statistics).

Item	Value
Frame sets	10,000 across five CARLA towns
Modalities per frame	RGB image, depth map, semantic segmentation image, single-channel class mask
Image files	40,000 (all AES-256 encrypted, with a SHA-256 manifest per town)
Native resolution	800 × 600 pixels (4:3)
Label format	YOLO: class id, normalised centre-x, centre-y, width, height
Classes	Car (includes vans, trucks, buses), pedestrian, cyclist (includes motorcycles)
Annotated objects	15,084 (11,008 cars, 2,424 pedestrians, 1,652 cyclists)
Frames without objects	32.6%; kept as valid background-only samples
Split (seed 42)	Train 7,000, validation 1,500, test 1,500

Frames without any detectable objects are denoted by empty label files. They are valid data samples and not errors and all scripts have been made to account for that. The only things needed for detector training are the RGB frames and labels.

3.2.2 Real world baseline: KITTI

The KITTI 2D Object Detection Benchmark [17] was selected for the true baseline since it doesn't depend on the statistics used to calibrate the generator (Waymo [18]). This eliminates the possibility of circular reasoning: any sim to real transfer performance will be due to generalization and not the statistical similarity that was created when calibrating the generator. The training image label pairs (7,481 in total) were reformatted into YOLO format.

Item	Value
Source	KITTI 2D Object Detection training split, 7,481 image–label pairs
Annotation conversion	Absolute pixel boxes (left, top, right, bottom) to normalised centre format
Car	Car, Van, Truck
Pedestrian	Pedestrian, Person_sitting

Item	Value
Cyclist	Cyclist
Excluded	Tram, Misc, DontCare
Split (70/15/15, seed 42)	Train 5,236, validation 1,122, test 1,123
Image size	About 1242 × 375 pixels (wide letterbox)

3.3 Experimental or study procedure

3.3.1 Dataset Description

The simulated dataset that is employed in the current experiment has been produced through the dataset generation module in the pipeline provided by the group through the CARLA simulation environment employing a privatized calibration setup. The dataset covers five different CARLA towns consisting of 10,000 annotated frame sets. Four different types of sensors are recorded for each individual frame including the following: an RGB image, a depth image, a semantic segmentation image and a class mask. All images are taken at 800 x 600 resolution.

The ground truth object annotations have been delivered in YOLO format where each annotation is one per detected object and consists of a class ID and the normalized centre – x, centre – y, width and height coordinates of the bounding box (normalized to the range [0,1]). There are three classes that have been labeled: car (covering vans, trucks and buses), pedestrians and cyclist (covering motorcycles). Frames containing no objects have been left empty and such frames will be treated as a proper data and all the scripts written for processing them should be able to cope with such an input.

For secure transfer, all the senior images (except for label files) have been encrypted using AES 256 with SHA 256 checksum for each town in order to check the integrity after decryption.

3.3.2 Environment Setup

A dedicated and isolated Python environment (Python 3.10 / 3.11) was constructed with the use of conda for the purposes of reproducibility. The following software packages have been installed: Ultralytics (with YOLOv8), PyCryptodome (for AES decoding), Pillow, Numpy, Pandas and Scipy (used for data processing and exploration), as well as torchmetrics with its extra modules for images (which provide an FID implementation). PyTorch has been installed with CUDA support where available. CUDA capability has been checked before starting the training process, which took place on CPU where there was no GPU driver.

3.3.3 Secure Decryption and Integrity Verification

All encrypted files consisted of an initialisation vector of 16 bytes and were then encrypted using AES 256 CBC with PKCS7 padding. A decryption pipeline was developed for the decryption of the original image data contained in all files in each of the five simulation environments. To prevent accidental disclosure of the key during deployment, the key was loaded at runtime from a managed configuration file instead of being hard coded in the source code, and the managed configuration file itself was not under version control.

Once decrypted, the SHA 256 hash for each file was computed and compared to the hash in the provided reference manifest of the dataset to identify if any file is corrupted and avoid their usage in future operations. The unencrypted annotation files (bounding boxes) were copied as-is, while the encrypted files were saved to allow decryption of all files in case of any future requirement without re acquiring the dataset. This approach led to 100% success rate of decryption, since all hashes matched the ones in the manifest.

3.3.4 Exploratory Data Analysis

The mode training was preceded by a statistical analysis of label distribution in the synthetic data set to better understand the structure of the latter and make predictions about factors that may influence future detection results. Among the total 10,000 labeled images, the imbalance among classes becomes obvious, with cars accounting for the majority of labeled objects (11,008 cars) and pedestrians (2,424) and cyclists (2,652) being represented to a lesser extent (the ratio being 5:1 between cars and pedestrians). Also, 32.6% of images had no labeled objects at all.

The analysis of the bounding box sizes found that pedestrians and cyclist objects are usually very small in size: 86.4% of pedestrian detections and 67.3% of cyclist detections have sizes less than 32x32 pixels, which has been demonstrated to be significantly harder to localize and detect using object detection models, such as YOLO. This result is methodological importance since, not taking it into consideration, one can erroneously assume that the low pedestrian and cyclist mAP is caused by problems with synthetic data generation instead of being a part of the object detection problem connected to the size of objects. The results per class are reported in this research.

3.3.5 Data Poisoning Detection

Since the data set had been through an external pipeline component, it was necessary to first validate the label files from a security standpoint before any training could begin, because there was a chance of data being poisoned, which is one case when it is possible to corrupt a model based on the annotations without changing the images in any way. Two types of validation were performed on each of the label files in the entire dataset (15, 084 annotated objects in five towns):

- Structural validation — checking each annotation line for the correct field count, a valid class identifier, normalised coordinate bounds (0 to 1), and a non-degenerate bounding box area.
- Statistical validation — comparing each town's per class label distribution against the distribution observed across the other towns, flagging any town whose class proportions deviated by more than four standard deviations, a pattern that would be expected if one town's labels had been selectively tampered with.

In order to ensure the efficient work of the detection algorithm while avoiding any simple answer like there are no anomalies, the performance of the detector was tested by adding two types of anomalies into an extra label file that is not included in the actual data set: out of range coordinates (x_center = 1.5) and a wrong class ID (class ID = 7).

3.3.6 Dataset Preparation for YOLO Training

Before conducting any form of training, the decrypted synthetic dataset comprising 10,000 image label pairs sampled from five simulation worlds was formatted following the required directory structure by the YOLOv8 training algorithm, which separated the images and labels into three different categories: train set (70% n = 7,000) , validation set (15%, n = 1,500) and test set (15%, n = 1,500). There being no distinct identifier among the five towns, it was necessary to rename the files by giving a prefix indicating the name of the town before moving them into the new directory structure to avoid having a conflict in names which would force us to eliminated a large part of the dataset. A seed value of 42 was selected in the randomization process to ensure the replicability of the split.

3.3.7 Model Training (Synthetic Baseline)

An object detection model (YOLOv8n, the smallest variant in the YOLOv8 family, comprising 3,006,233 parameters) was trained on the synthetic training split (7,000 images) for a maximum of 50 epochs, at an input resolution of 640×640 pixels, with a batch size of 8 and a fixed random seed (seed = 42) for reproducibility. Training was conducted on CPU (Intel Core i5-8250U) due to the absence of a compatible GPU on the available hardware; this increased training time substantially (approximately 50× slower than GPU execution) but did not affect the correctness of the resulting model, since all comparisons in this evaluation were conducted under identical hardware conditions. An early stopping criterion (patience = 15 epochs without validation improvement) was applied to avoid unnecessary computation once performance plateaued, and the best performing checkpoint (by validation mAP) was retained for subsequent evaluation.

3.3.8 mAP Evaluation Methodology

The model's performance was assessed based on a particular test set that is distinct from the validation set during the procedure of determining the criteria for stopping the trained process to prevent any bias caused by validation leakage. Two types of mean average precision metrics were assessed, which include mAP50 (when the overlap is at least 50% between the ground truth bounding box and the predicted bounding box), and mAP50-95 (average of mAPs from 50% to 95%). The stricter version is selected to assess the model's performance because it considers localization capability compared to mAP50.

3.3.9 Real Data Baseline: Selection and Preparation (KITTI)

Synthetic mAP alone provides little indication of practical utility without a real world point of comparison. The KITTI 2D Object Detection Benchmark was selected as an independent real world dataset for this purpose. KITTI was chosen deliberately as a dataset independent of the statistical sources used to calibrate the synthetic data generation pipeline (Stage 1, which draws on Waymo Open Dataset statistics), thereby avoiding circularity between the calibration source and the evaluation baseline and ensuring that any measured sim to real transfer reflects genuine generalization rather than statistical similarity introduced during privacy calibration.

The KITTI training split (7,481 image-label pairs) was downloaded and converted from KITTI's native annotation format to YOLO format. KITTI's absolute pixel coordinate bounding boxes (left, top, right and bottom) were converted to normalised Centre coordinate format, consistent with the label format of the synthetic dataset. Class mapping followed the same three class taxonomy used throughout the project: Car, Van and Truck were mapped to car; Pedestrian and Person_sitting were mapped to pedestrian; Cyclist was mapped to cyclist. The Tram, Misc and DontCare categories were excluded, consistent with the synthetic dataset's class structure. The converted dataset was split into train/validation/test sets using the same 70/15/15 ratio and fixed random seed (42) used for the synthetic dataset, yielding 5,236 training, 1,122 validation, and 1,123 test images.

A notable limitation of using KITTI as the real world baseline is the substantial difference in image resolution and aspect ratio between the two domains: CARLA generated synthetic images are 800×600 (4:3 aspect ratio), while KITTI images are approximately 1242×375 (a wide letterbox format). This is acknowledged explicitly as a confounder of the sim to real comparison: part of any observed performance gap between synthetic trained and real trained models may originate from this resolution and aspect ratio mismatch, rather than exclusively from the privacy preserving synthetic-data generation process itself.

3.3.10 Cross-Domain Evaluation Protocol

To determine how much utility is lost in terms of the privacy preserving synthetic data compared to real world data, a method involving four experiments was devised. Two separate YOLOv8 object detection models were trained; the first one using the synthetic data obtained from CARLA and the second one using real world data obtained from KITTI. The two models were then tested on the testing dataset from their respective domains as well as on the testing dataset from the other domain, resulting in four train / test pair combinations as described below.

#	Train Domain	Test Domain	Purpose
1	Synthetic	Synthetic	Can the model learn from this data at all?
2	Synthetic	Real (KITTI)	Sim to real transfer the primary utility metric
3	Real (KITTI)	Real (KITTI)	Upper bound with real data
4	Real (KITTI)	Synthetic	Reverse check on the domain gap

Reproducibility Information: Reproducibility information: model architecture – YOLOv8n; framework – Ultralytics 8.4.115; evaluation metrics – mAP50 (IoU ≥ 0.5), mAP50-95 (average IoU 0.5 – 0.95); test set size – synthetic 1,500 images, KITTI 1,123 images; random seed – 42; hardware – CPU (Intel i5-8250U).

3.3.11 FID Protocol

To complement the task level utility measurement, the Fréchet Inception Distance (FID) was computed to assess the image level statistical similarity between the synthetic (CARLA generated) and real world (KITTI) datasets. FID measures the distance between the feature distributions of two image sets, extracted using a pre-trained InceptionV3 network (2048 dimensional feature space), with lower values indicating greater statistical similarity. The feature extractor used was InceptionV3 (via the torchmetrics FrechetInceptionDistance implementation); images were resized to 299×299 and normalised prior to feature extraction. A sample size of 1,000 images per set (real and synthetic) was used, meeting the minimum threshold

recommended for reliable FID estimation; the real-image source was the KITTI test split and the synthetic-image source was the synthetic test split.

3.4 Measures and Evaluation Metrics

In summary, this evaluation combines four complementary forms of measurement: **(i)** a security audit (data poisoning detection, including an empirical detector validation test); **(ii)** task level detection performance (mAP50 and mAP50-95, overall and per class); **(iii)** cross domain generalization (the four experiment protocol in Section 3.12); and **(iv)** distributional realism (FID). Together, these measurements constitute a complete utility evaluation that directly addresses the research aim stated.

Measure	Definition	Purpose
mAP50	Mean AP over classes at IoU ≥ 0.5	Lenient detection accuracy
mAP50-95	Mean AP averaged over IoU 0.5 to 0.95	Primary metric; rewards precise localisation
Per-class AP50 and AP50-95	AP for car, pedestrian and cyclist	Exposes class-level failure hidden by aggregates
Utility retention	Synthetic-trained mAP on real data divided by real-trained mAP on real data	Size of the sim-to-real gap
FID	Fréchet distance between Inception-V3 features of the two image sets	Image-level realism
Poisoning anomalies	Structural and statistical flags per label set	Label integrity
Decryption success	Files whose SHA-256 hash matches the manifest	Transfer integrity

mAP 50-95 is the primary metric because it reflects localization accuracy better than mAP50 alone. Per class values are reported alongside aggregates so that rare classes are not hidden.

3.5 Data analysis

The formula for calculating utility retention was ((synthetic -> real mAP) / (real -> real mAP)) * 100 , for mAP50, mAP50-95 and per class. The logical consistency of ranking (real -> real > synthetic -> synthetic > both cross – domain) was examined among the four experiments as an indicator of internal validity. A large gap between the two cross domain numbers would indicate the presence of a one sided artifact from the dataset, hence the reverse experiment (real -> synthetic) was conduct to check whether the difference is mutual. FID was assessed using the commonly used scale (FID below 50 reasonable similarity, 50 – 200 moderate difference, above 200 very different), and was contrasted with the retention results to test whether realism predict utilty. This is a rule of thumb, not a statistical test. The poisoning audit relied on the structural rules and four standard deviation rule.
Each model was trained only once, with seed 42, hence the results are single run numbers with no confidence intervals. This is addressed as a weakness.

4. Product and Solution Design

4.1 Introduction

This chapter is a description of the pipeline that was created as a tool to evaluate the utility of this particular component. During the implementation process, the method described in Chapter 3 was implemented as an actual set of Python scripts that can decrypt and validate the dataset, explore it, perform an audit, train and evaluate object detection models and measure utility.

4.2 Requirements

The requirements below were derived from the research aim in Section 1.5 and from the constraints of receiving data from an external pipeline component.

Functional requirements

- **FR1.** Decrypt every encrypted file (16-byte IV followed by AES-256-CBC ciphertext with PKCS7 padding) across all five towns.
- **FR2.** Verify the SHA-256 hash of every decrypted file against the town manifest and report any mismatch.
- **FR3.** Handle empty label files (background-only frames) as valid data without failure.
- **FR4.** Compute per-class counts, per-frame counts, the proportion of empty frames and the bounding-box size distribution.
- **FR5.** Validate every label file structurally and statistically, report a PASS/FAIL verdict, and demonstrably detect deliberately injected defects.
- **FR6.** Convert the synthetic dataset and the KITTI dataset to the same YOLO structure using the same 70/15/15 split and seed (42), with town-prefixed file names to avoid collisions.
- **FR7.** Train a synthetic-domain model and a real-domain model with identical hyperparameters.
- **FR8.** Evaluate the four train/test combinations on held-out test splits, reporting overall and per-class mAP50 and mAP50-95.
- **FR9.** Compute FID between 1,000 real and 1,000 synthetic images.

Non – functional requirements

- **Reproducibility.** Fixed seeds, recorded software versions and documented hyper parameters, so that all results can be regenerated.

- **Security.** The decryption key must not appear in source code or version control, and encrypted originals must be retained.
- **Integrity.** Decryption must be lossless and verifiable, and the label data must be audited before it is used for training.
- **Robustness.** Scripts must not crash on empty labels or on differences in annotation format.
- **Comparability.** Both domains must be processed with identical splits, hyper parameters and hardware.
- **Independence.** The real-world baseline must be independent of the data used to calibrate the generator.
- **Feasibility.** All steps must run on the available CPU-only hardware.
- **Transparency.** Per-class results and known confounders must be reported rather than hidden behind aggregate figures.

4.3 Component architecture within the complete solution

This component occupies the third stage of the group's four stage pipeline (Section 1.1). It consumes the encrypted synthetic dataset produced by the dataset generation component (Stage 2) and produces, as its output, a set of quantitative utility findings (mAP, FID, security audit results) that feed directly into the group's combined privacy utility argument, alongside the model level privacy cost findings produced by the privacy calibration component (Stage 1).

4.4 Design alternatives and decisions

Table 2 summarizes the main design alternatives that were considered and the decisions taken for this component.

Decision area	Alternative considered	Decision and rationale
Real-world baseline	Reuse the Waymo Open Dataset used to calibrate the generator	KITTI was chosen. An independent dataset avoids circularity between the calibration source and the evaluation baseline.
Experiment design	Only synthetic-to-real transfer	Four train/test combinations were used. They add a learnability check, a real-data upper bound and a reverse check of the domain gap.
Reporting level	Aggregate mAP only	Per-class and per-size reporting was adopted. Aggregate mAP would mask the weak pedestrian and cyclist results. mAP50-95 is the primary metric.
Evaluation split	Report on the validation split	A separate held-out test split is used, so early-stopping decisions cannot leak into the reported results.
Poisoning detection	Structural checks only	Structural and statistical (four standard deviations between towns) checks were combined, and the detector was proven with deliberately injected defects.
Detector model	Larger YOLOv8 variants	YOLOv8n (3,006,233 parameters) was used because training had to run on a CPU. Both domains use identical hyperparameters.
Realism metric	mAP only	FID was added as a complementary, task-independent measure, using 1,000 images per set.

Decision area	Alternative considered	Decision and rationale
Privacy budgets	Evaluate several epsilon values	Not possible with the available data, because epsilon is applied upstream of scene generation. A single configuration (epsilon = 1.8) is evaluated and the protocol is reusable.
Empty-label frames	Exclude them from the dataset	They are retained as valid background-only samples (32.6% of frames), and the scripts handle them explicitly.
File naming	Keep original frame identifiers	Town-prefixed names were used. Frame indices repeat across towns, so original names would silently overwrite files.
Key management	Hard-code the key in the scripts	The key is read at runtime from an environment-managed configuration file that is excluded from version control.

Table 2: Design Alternatives and Decision

4.5 Detailed design

4.5.1 Decryption Module

The decryption module (decrypt.py) implements AES 256 CBC decryption with PKCS7 padding removal, reading the 32 byte key from an environment variable rather than hardcoding it in source code. For every encrypted file across the five town directories, the module decrypts the file, computes its SHA 256 hash, and compares the hash against the town's reference manifest, reporting any mismatch immediately. Non encrypted label files are copied without modification. The module was designed so that encrypted originals are never deleted, allowing the decryption process to be safely repeated without re requesting the dataset.

4.5.2 Data Exploration Module

The exploration module iterates over every label file in the decrypted dataset, computing per class object counts, per frame object counts, the proportion of empty (background only) frames, and the pixel area distribution of each class relative to the native 800×600 image resolution. These statistics informed the class imbalance and small object findings reported in Section 3.6, and directly justify per class and per size reporting adopted throughout this evaluation.

4.5.3 Poisoning Detection Module

The poisoning detection module (poison_detection.py) performs the structural and statistical validation described in Section 3.7 across every label file in the dataset, producing a per town class distribution summary, a list of flagged anomalies (if any), and an overall PASS/FAIL verdict. A companion script (prove_poison_detection.py) deliberately introduces two known defects into an isolated copy of a label file and re runs the detector against it, providing empirical evidence that the detector actively identifies tampering rather than passing all input unconditionally.

4.5.4 Dataset Conversion Module

Two conversion scripts were implemented. The first converts the decrypted synthetic dataset into the directory structure and data.yaml configuration required by the Ultralytics YOLOv8 framework, applying the town prefixed renaming and fixed seed 70/15/15 split described in Section 3.8. The second converts the downloaded KITTI dataset whose native annotation format uses absolute pixel coordinates in a different field order into the same YOLO format structure, applying the class mapping described in Section 3.11 and the identical 70/15/15 split and seed, ensuring structural consistency between the synthetic and real world datasets for the downstream cross domain evaluation.

4.5.5 Model Training Module

Two independent training runs were executed using the Ultralytics YOLO training API: one on the synthetic dataset (producing the synthetic_baseline checkpoint) and one on the converted KITTI dataset (producing the kitti_baseline checkpoint), using identical hyper parameters (50 epochs, image size 640, batch size 8, seed 42, early-stopping patience 15) to isolate the effect of the training data domain rather than training-configuration differences.

4.5.6 Cross Domain Evaluation Module

The cross evaluation module (cross_evaluation.py) loads both trained model checkpoints and iterates over the four train/test combinations described in Section 3.12, invoking the Ultralytics validation API against the relevant dataset configuration in each case and recording overall and per class mAP50 and mAP50-95 for each experiment, consolidated into the summary table presented in Chapter 5.

4.5.7 FID Computation Module

The FID module (compute_fid.py) loads 1,000 images from each of the real (KITTI) and synthetic test splits, resizes and normalises them, and streams them in batches through the torchmetrics Fréchet Inception Distance implementation (InceptionV3 backbone, 2048 dimensional feature space), reporting the final computed FID score alongside the number of images used per set, in line with the reporting requirement for small sample FID reliability.

4.6 Standards and professional practice

The following standards and professional practices were followed.

- **Cryptography.** AES – 256 as per NIST FIPS 197 [8] was applied for decryption key was removed from source code and version control.
- **Data Protection.** Motivation and design are aligned with GDPR [11] and the Personal Data Protection Act of Sri Lanka [12]. Evaluation uses synthetic imagery, and only real world data that is being used is the publicly available KITTI benchmark with license terms for academic research purposes.
- **Privacy.** The theoretical background of differential privacy is based on the work of Dwork and Roth [1]. This component uses the privatized output of stage 1 and does not degrade its quality.
- **Security practices.** Data provided by another component is considered untrusted and inspected for poisoning [9]. Poisoning detector is tested using a test with known defect and encrypted original images are stored to make further processing repeatable.
- **Academic integrity.** Generative AI use is disclosed in Appendix B in accordance with the module policy.

5 Implementation Integration and Testing

5.1 Development environment and technology stack

All methods related to data processing, security checks and model training, as presented in this study, have been done in Python (3.10 / 3.11) in a separate conda environment. The object detection model training and prediction tasks have been executed using the Ultralytics YOLOv8 framework, based on PyTorch deep learning library. Decryption of the cipher text has been done through the usage of PyCryptodome library whereas statistical computations and data exploration have been done with the help of Numpy, Scipy and Pandas.

Because of hardware limitations particularly, lack of a suitable GPU driver for the development machine, the model was trained using CPU (Intel Core i5-8250U) instead of GPU. Although this did not impact the accuracy of the output, it significantly increased the training time compared to training with GPU acceleration (about 50 times slower).

5.2 Component Implementation

Several practical challenges were encountered and resolved during implementation:

- As a consequence of the Ultralytics default project folder generation, the trained model checkpoint files needed to be explicitly checked within the file system in order to make sure that the right paths were used in the cross evaluation script.
- While preparing the datasets, care had to be taken to take into account those image labels which were empty in order not to crash the script, because 32.6% of synthetic frames and 475 out of 1,500 synthetic test images contained no objects at all.
- Converting KITTI's native annotation format (absolute pixel coordinates in a non-YOLO field order) to the normalised YOLO format required careful coordinate transformation logic and class mapping, with validation against the dataset's documented format.
- The downloading, registration and extraction of the KITTI benchmark dataset (approximately 12 GB of images) within the set limits of storage space and time necessitated the proper handling of interim raw data files.
- Machine learning training of all models used took place using CPU only as there was no GPU available, which implied overnight training sessions and taking special care of the computer’s sleep mode settings.

5.3 Integration with the other components

This element is both the consumer of the data generation stage and the producer of the information needed by other stages.

Connection to stage 2 (Dataset Generation). The data generation element produces encrypted by AES – 256 images (RGB, depth, semantic segmentation and class mask at 800 x 600 resolutions), plain text YOLO label files and SHA – 256 manifests for each of the five towns. These are consumed by the decryption element, and the manifest comparison showed that all 40,000 images arrived without issues. Of these, only the RGB images and labels are needed for the detector training. As frame IDs overlap between towns, town based renames were made at the receiving end in order to incorporate five towns into one dataset.

Connection to stage 1 (Privacy Calibration). The synthetic data was created based on privatized with Laplace mechanism parameters at epsilon = 1.8; that is the reason why all the results depend on this particular parameterization. This way, the model level cost found in stage 1 (DP – SGD: 0.17 AUC loss, with AUC loss of membership inference attack from 0.65 to 0.50) connects with the data level cost found here table 5 (Connecting Findings to the Group Pipeline).

Interface with stage 4 (Risk and Compliance). The verdict of poisoning audit (PASS, no anomalies found out of 15,084 boxes), the encryption and key management controls and the

results of integrity verification can be used as proof of application of controls for the risk and compliance module at the handover of Stages 2 and 3.

Integration Testing. The integration was checked by means of decryption test (40,000 out of 40,000 files, zero errors), conversion test (7,000 / 1,500 / 1,500 split synthetics) and successful completion of both training and all four evaluation runs. If more datasets are generated by the generation stage for different epsilons, they could pass the same modules untouched.

5.4 Verification and Software Testing

Each functional requirement was verified with the full experiment.

ID	Requirement	Verification	Status
FR1	Decrypt all encrypted files	40,000 of 40,000 decrypted, 0 failed	Passed
FR2	Verify SHA-256 against manifests	All computed hashes matched	Passed
FR3	Handle empty labels	475 of 1,500 test images are empty; processed without error	Passed
FR4	Dataset statistics	Class counts, 32.6% empty frames and size shares computed	Passed
FR5	Poisoning audit and detector proof	PASS on 15,084 boxes; injected defects flagged (3 anomalies)	Passed
FR6	Conversion and split	7,000/1,500/1,500 synthetic; 5,236/1,122/1,123 KITTI (7,481 converted)	Passed
FR7	Identical hyper-parameters for both models	Both trained with 50 epochs, size 640, batch 8, seed 42, patience 15	Passed
FR8	Four-experiment evaluation	All four runs completed; ranking Real→Real > Synthetic→Synthetic > cross-domain pairs	Passed
FR9	FID on 1,000 images per set	FID = 115.99	Passed

6 Testing and Evaluation

6.1 Testing Procedure

This chapter reports the results of applying the Chapter 3 methodology: security audit, synthetic only evaluations, real data baseline, cross domain evaluation and FID computation.

6.2 Poisoning Detection Results

The structural and statistical checks on the full dataset (15,084 object instances, five towns) found zero anomalies and no significant between town class distribution deviation, giving a PASS verdict. A deliberately poisoned label file (x_center = 1.5, class ID 7) was correctly flagged (3 anomalies), confirming that the detector actively identifies tampering and that the dataset was not corrupted before training.

6.3 Synthetic Data Model Performance

On the held out test subset (1,500 images, 2,367 instances), excluded from training and validation, the synthetic baseline achieved mAP50 = 0.3655 and mAP50-95 = 0.2273. Per class AP50-95 was 0.3981 (Car), 0.1882 (Cyclist) and 0.0955 (Pedestrian); Car performance was about 4.2 times that of Pedestrian, reflecting the roughly 5:1 class imbalance and small object sizes found in the exploratory analysis.

6.4 Real Data Baseline (KITTI) Preparation Results

KITTI conversion produced 7,481 image label pairs (5,236 train / 1,122 validation / 1,123 test; 70/15/15, seed = 42) in the same layout as the synthetic dataset. A second YOLOv8n model (kitti_baseline) was trained with identical hyper parameters, isolating the effect of the training data domain.

6.5 Cross Domain Evaluation Results

Table 3 summarizes the four experiments cross domain protocol described in Section 3.12.

Experiment	mAP50	mAP50-95	Car AP50	Pedestrian AP50	Cyclist AP50
Synthetic → Synthetic	0.3655	0.2273	0.6076	0.1793	0.3096
Synthetic → Real (KITTI)	0.0889	0.0343	0.2614	0.0051	0.0001
Real (KITTI) → Real (KITTI)	0.8500	0.5531	0.9519	0.7769	0.8213
Real (KITTI) → Synthetic	0.0507	0.0199	0.1273	0.0079	0.0171

Table 3: Full Cross-Domain Evaluation Results

Synthetic-to-real mAP50 (0.0889) represents 10.46% of the real-to-real mAP50 (0.8500), that is, (0.0889 ÷ 0.8500) x 100, the quantifiable price of privacy on the data side (Section 1.3). However, the cost distribution is not even: while Car kept about 27.5% of real data AP50 (0.2614 vs. 0.9519), Pedestrian (0.0051) and Cyclist (0.0001) dropped, reflecting class imbalance and small objects (Section 3.6). Conversely, the reverse transfer (Real → Synthetic, mAP50 = 0.0507) indicates that the gap between domains goes both ways (Section 3.11), and the consistent ordering Real → Real > Synthetic → Synthetic > cross domain pairs underpins protocol internal validity.

6.6 FID Results

The computed FID between the synthetic and real (KITTI) test image sets was 115.99 (n = 1,000 images per set).

Range	Interpretation
< 50	Reasonable similarity
50 – 200	Moderate difference
> 200	Very different

Table 4: FID Interpretation against Reference Scale

The FID between the synthetic and real (KITTI) test sets was 115.99 (n = 1,000 per set), within the ―moderate difference‖ range (< 50 reasonable similarity, 50–200 moderate, > 200 very different), yet this did not translate into proportional utility (10.46% mAP retention, near zero Pedestrian and Cyclist AP). FID alone therefore does not predict object level detection performance; mAP and FID should be used together.

6.7 Consolidated Testing Summary

Test Item	Expected Result	Actual Result	Status
Dataset decryption	All files decrypted	40,000/40,000, 0 failed	Passed
Poisoning audit and deliberate test	Clean passes; defects flagged	PASS; 3 anomalies flagged	Passed
YOLO and KITTI conversion	70/15/15 split	7,000/1,500/1,500; 7,481 converted	Passed
Training and cross-domain evaluation	Runs complete; logical mAP order	All completed; order as expected	Passed
FID computation	≥ 1,000 images per set	FID = 115.99	Passed

Table 5: Functional and Experimental Test Results

6.8 Deployment and Operation

The solution itself is implemented as a chain of batch style evaluation programs and is not running as a service. The chain should be executed in the following manner:

1. The decryption key should be set in the configuration file managed by the environment (and never committed to version control), and the decrypted.py program should be executed to restore the images and check SHA 256 hashes.
2. The exploration program to obtain class, empty frame and object size statistics should be run.
3. The poison detection program (poison_detection.py) and to make sure it works the companion prove_poison_detection.py program should be run.
4. The two scripts for the creation of the synthetic and KITTI dataset in YOLO format in 70/15/15 distribution with seed 42 should be executed.
5. The synthetic baseline and kitti_baseline should be trained using exactly the same parameters (50 epoches, images size 640, batch size 8 and patience 15).
6. The cross domain evaluation (cross_evaluation.py) and FID computation (compute_fid.py).

6.9 Discussion of Results

The synthetic dataset trains a functional detector (mAP50=0.3655) and all tests passed, but utility transfers poorly to real data (10.45% retention), mainly for Pedestrian and Cyclist. Two confounders apply: the CARLA (800x600) vs. KITTI (1242 x 375) resolution and aspect ratio mismatch, so part of the drop is not due to privacy preserving generation itself and cannot be fully isolated; and CPU only training, which increased run time but not validity since all comparisons used identical hardware.

7 Discussion

7.1 Interpretation of findings

The synthetic dataset trains a functional detector (mAP50=0.3655) and all tests passed, but utility transfers poorly to real data (10.45% retention), mainly for Pedestrian and Cyclist. Two confounders apply: the CARLA (800x600) vs. KITTI (1242 x 375) resolution and aspect ratio mismatch, so part of the drop is not due to privacy preserving generation itself and cannot be fully isolated; and CPU only training, which increased run time but not validity since all comparisons used identical hardware.

7.2 Research contribution

This part (i) measures the cost of privacy (10.46% retention) in terms of data; (ii) establishes a relationship between the pedestrian and cyclist collapse and class imbalance and small objects; (iii) proves that there is a discrepancy between FID and mAP; (iv) employs an independent methodology, in which the KITTI is separate from the Waymo calibration source and there are four experiments, among which are an upper bound on the real data and a reverse transfer; (v) introduces a protocol applicable to other epsilon values; and (vi) provides a label integrity audit.

7.3 Product contribution

A reproducible set of Python modules (decryption and integrity, exploration, poisoning detection, conversion, training, cross-domain evaluation and FID; Section 4.4) that considers the issue of empty labels, avoids naming conflicts, hides secrets from version control and allows repeating on any future dataset of the same format.

7.4 Limitations and threats to validity

There is only one privacy parameter (epsilon = 1.8), so the privacy–utility curve cannot be constructed; the KITTI–CARLA resolution and aspect-ratio difference is an uncontrolled confounder; all the experiments were run on CPU.

7.4.1 Internal Validity

- Each model was trained once (seed 42), so the results have no confidence intervals. Differences such as 0.0889 against 0.0507 should be read as indicative, whereas the large gaps are far beyond plausible seed noise.
- The KITTI split was random over single frames. KITTI frames come from continuous drives, so near-identical frames can appear in both training and test splits, which probably raises the real-to-real result (0.8500). A sequence-aware split would give a more conservative upper bound.
- Early stopping used a validation split, and the test split was used only for reporting, which prevents validation leakage. The 50-epoch limit and the small YOLOv8n model, chosen because of CPU-only training, may keep both models below their best possible accuracy.

7.4.2 Construct Validity

- The 10.46% figure combines the privacy effect with the rendering, resolution and aspect-ratio gap and the class-taxonomy differences. No synthetic dataset calibrated without noise was available, so the true cost of privacy cannot be isolated.
- The class mapping is only approximately equivalent: CARLA cars include buses and trucks, CARLA cyclists include motorcycles, and KITTI excludes tram, misc and don't-care objects.
- Per-object-size mAP was not computed. Size is analysed through the label statistics (86.4% of pedestrians and 67.3% of cyclists below 32×32 pixels), so the size explanation is supported by the data composition but was not tested by a separate size-stratified evaluation.
- FID used 1,000 images per set and a 299×299 resize that distorts the two aspect ratios differently, so part of the FID reflects image shape and not content. The 50/200 reference scale is a rule of thumb.
- The poisoning audit finds structural defects and town-level distribution shifts. It cannot detect a small number of subtle, internally consistent label changes.

7.4.3 External Validity

- The results come from one simulator, one calibration setting, one detector (YOLOv8n), one real benchmark (KITTI) and three classes. They may not hold for other detectors, simulators or privacy budgets.

7.5 Implications

There is a quantifiable cost to privacy at both data and model levels (Table 5). So statements must include both aspects. Generators should be evaluated based on their capacity to faithfully represent the vulnerable road users and this dataset by itself is not fit for deployment. The FID is not enough to prove the utility of the dataset; a task-level evaluation on actual data is required per class. 10.46% includes the gap between the rendering and resolution discrepancy and the privatisation effect, thus it is an upper bound on the cost of privacy when epsilon = 1.8.

8 Conclusion

8.1 Conclusion

This work aimed at empirically evaluating whether the privacy preserving synthetic driving dataset, created using a CARLA simulator which was calibrated by differential privacy parameters, is still useful enough to train an object detection model for autonomous vehicles as compared to the real world dataset. It has been carried out using an entire utility evaluation pipeline consisting of decryption and validation of data, data exploration and detection of poisoned data and its empirical validation, training the YOLOv8 model, conducting four experiments on cross domain evaluation and calculating Fréchet Inception Distance. This module has helped in completing the privacy utility trade off of the research work carried out by the group.

The results show that the synthetic dataset is capable of training a functional perception model (mAP50 = 0.3655 on synthetic data), but retains only 10.46% of real data detection performance when transferred to real world imagery, with this utility cost concentrated disproportionately on the pedestrian and cyclist classes due to class imbalance and small object prevalence in the dataset. The data poisoning audit confirmed, with empirical validation, that the received dataset was free of tampering, and the FID score of 115.99 indicated moderate, rather than extreme, distributional dissimilarity between synthetic and real imagery a finding that, combined with the mAP results, demonstrates that aggregate visual realism alone does not guarantee object detection utility.

This component plays an important role in the broader group project, as it provides the data level cost measurement that completes the project's central privacy utility argument, pairing directly with the model level cost (DP – SGD) quantified by the privacy calibration component.

8.2 Achievement of Objectives

Objective	Outcome	Status
1. Decrypt and verify the dataset	40,000 of 40,000 files decrypted; all SHA-256 hashes matched	Achieved
2. Explore class and size composition	5:1 car-to-pedestrian ratio, 32.6% empty frames, 86.4% / 67.3% of pedestrians / cyclists below 32×32 pixels	Achieved
3. Validate a poisoning detector	PASS on 15,084 boxes; injected defects flagged	Achieved
4. Train YOLOv8n and report overall and per-class mAP	mAP50 = 0.3655, mAP50-95 = 0.2273; per-class results in Table 6.2	Achieved

Objective	Outcome	Status
5. Real baseline and four-experiment evaluation	KITTI baseline and four experiments completed; 10.46% retention	Achieved
6. Compute FID	FID = 115.99 (n = 1,000 per set)	Achieved
Group target: utility within 15% of real data	10.46% of the real-data mAP50	Not reached

8.3 Individual research and Product contributions

The individual research contributions are:

- A measured, per class estimate of the data level utility of a differentially private CARLA dataset against an independent real benchmark.
- Evidence that FID doesn't predict detection utility, which supports using task level evaluation next to realism metrics.
- A four experiment protocol, including a real data upper bound and a reverse transfer that can be applied to datasets at other ε values.
- A label integrity audit validated with injected defects.

The individual product contributions are:

- The set of reproducible evaluation modules
- The secure decryption and integrity pipeline with runtime key handling
- The converters that put the synthetic and KITTI data in an identical YOLO structure
- The clean labels and utility results that connect to the other components

8.4 Limitations and Future Improvement

Even though all objectives have been met, there are certain limitations associated with this evaluation. Specifically, the synthetic dataset has been created using only one value for the privacy parameter (epsilon=1.8). This is because of the fact that in the existing pipeline design, the privatization of configuration parameters occurs upstream from the scene generation and not a dataset itself. Therefore, it was impossible to construct a privacy utility curve as part of the current research. The real world dataset (KITTI) differs significantly from the synthetic dataset in terms of resolution and aspect ratio. This creates another confounding variable that could not

be controlled. All experiments have been carried out using the CPU as the GPU was not available.

In terms of future improvements for this research, in case of running the dataset creation process for multiple calibration epsilon values, the methodology introduced in this report could be simply replicated for each generated dataset in order to build up a privacy versus utility trade off curve.

An alternative real world baseline with similar aspect ratio like the Cityscapes dataset would help in reducing the problem with resolution mentioned in section 3.11. Domain adaptation methods, including fine tuning of the synthetically trained model on a few real world samples, could be also tested as an appropriate way of closing the simulation real gap, together with GPU acceleration training.

8.5 Final Remarks

This component's findings integrate with the group's broader investigation into the privacy utility trade off of the proposed pipeline, as summarized in Table 6.

Level	**Component**	**Finding**
Data	Utility Evaluation (this work)	Synthetic data retains 10.46% of real data detection utility.
Model	Privacy Calibration (DP-SGD)	DP-SGD costs 0.17 AUC and defeats membership inference (attack AUC 0.65 → 0.50).

Table 6: Connecting Findings to the Group Pipeline

Both these points together prove that there is an observable price to be paid in terms of utility in the processes of data creation and model training by the application of privacy preserving mechanisms, thus proving the point made in the project that privacy preserving techniques have a cost associated with them and this element has measured it.

9 References

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Glossary

Average Precision (AP). The area under the precision recall curve of one class.

Class imbalance. A dataset property in which some classes have many more example than others.

Domain gap. The loss of accuracy when a model is applied to data from a different distribution than it was trained on.

Fréchet Inception Distance (FID). A distance between the Inception – V3 feature distributions of two image sets; lower means more similar

Data poisoning. Tampering with training data or labels so that a trained model behaves wrongly.

Sim-to-real gap. The loss in performance when a model trained on simulated data is applied to real data.

Utility retention. The synthetic-trained model's score on real data as a percentage of the real-trained model's score on real data.

YOLO label format. One line per object: class id followed by normalised centre-x, centre-y, width and height.

Intersection over Union (IoU). The overlap between a predicted and a ground truth box divided by their union.

mAP50 / mAP50-95. Mean AP at an IoU threshold of 0.5 / averaged over thresholds from 0.5 to 0.95.

Appendices

Appendix A Extended results

Item	Value
Total annotated objects	15,084
Cars	11,008 (73.0%)
Pedestrians	2,424 (16.1%); 86.4% smaller than 32×32 pixels
Cyclists	1,652 (11.0%); 67.3% smaller than 32×32 pixels
Frames with no annotated object	32.6% of 10,000
Synthetic test split	1,500 images, 2,367 instances, 475 empty images
KITTI test split	1,123 images

Quantity	Calculation	Result
mAP50 retention	0.0889 ÷ 0.8500	10.46%
mAP50-95 retention	0.0343 ÷ 0.5531	6.20%
Car AP50 retention	0.2614 ÷ 0.9519	27.5%
Pedestrian AP50 retention	0.0051 ÷ 0.7769	0.66%
Cyclist AP50 retention	0.0001 ÷ 0.8213	0.01%
Real → Synthetic relative to Synthetic → Synthetic (mAP50)	0.0507 ÷ 0.3655	13.9%
Car to pedestrian AP50-95 (synthetic test)	0.3981 ÷ 0.0955	4.2×

Script	Function
decrypt.py	AES-256-CBC decryption and SHA-256 manifest verification
Exploration script	Class, size and empty-frame statistics
poison_detection.py	Structural and statistical label audit with PASS/FAIL verdict
prove_poison_detection.py	Injects known defects into an isolated copy to validate the detector
Synthetic converter	YOLO directory structure, data.yaml, town-prefixed names, seeded 70/15/15 split

Script	Function
KITTI converter	KITTI-to-YOLO conversion, class mapping, same split and seed
Training (Ultralytics)	synthetic_baseline and kitti_baseline checkpoints
cross_evaluation.py	Four-experiment evaluation with overall and per-class mAP
compute_fid.py	FID on 1,000 images per set

Appendix B Academic integrity and AI use disclosure

- Generative AI was used while preparing this these, in line with the module policy on AI use.
- Tool: - Claude, a large language model by Anthropic
- Use: - The tool helped to restructure the author's draft into the final chapter structure, to draft and edit text from the author's own research paper, results and draft thesis, and to cross check numbers between documents.